



# MedGuard RAG

Drug Interaction & Dosage Clinical Decision Support

Retrieval-Augmented Generation

Evidence-Grounded, Cited Answers

Helps clinicians check whether two drugs interact dangerously, what the severity is, and what FDA-documented warnings or dosage guidance apply — with every answer traced back to a real source, never an unverified model guess.

## WITHOUT RETRIEVAL

A plain LLM answers drug-interaction questions from memory alone — it can **hallucinate** interactions, severities, or dosages that sound plausible but aren't real. In healthcare, a confident wrong answer is a patient-safety risk.

## WITH MEDGUARD RAG

Real evidence is retrieved first — from a structured interaction database and real FDA drug labels — **before** any answer is generated. Every claim is cited; uncertainty is flagged instead of guessed.

## DATA SOURCES — REAL, NOT SYNTHETIC



### DDInter 2.0

Academic drug–drug interaction database (blood / blood-forming organs category)

15,140 records 1,503 drugs 4 severity tiers



### openFDA

Official FDA label text: boxed warnings, contraindications, dosage, interaction sections

Single-ingredient drugs Live API

## SYSTEM ARCHITECTURE

1

### Data Ingestion Structured + Unstructured

DDInter CSV loaded into a SQL database. FDA label text fetched via the openFDA API, stripped of HTML markup, and split into small, sentence-aware retrievable chunks.

CSV → SQL

HTML cleaning

sentence-aware chunking

2

### Dual Storage Structured facts + Vector index

PostgreSQL / SQLite holds structured interaction facts plus chunk text and metadata. Qdrant holds numerical embeddings of each chunk for meaning-based search.

PostgreSQL / SQLite

Qdrant vector DB

embeddings

3

### Hybrid Retrieval Exact lookup + Semantic search

Combines exact structured lookups (drug pair → severity) with semantic vector search that finds FDA text matching the meaning of a question, even with different wording.

drug-pair lookup

semantic search

result fusion

4

### Generation Claude API

Claude produces the final answer strictly grounded in retrieved evidence — always citing sources, always flagging uncertainty rather than guessing.

Claude API

source citation

uncertainty flagging

5

### Serving API + UI

FastAPI backend and an MCP server expose the pipeline; a Streamlit UI provides a clinician-facing demo for interactive use.

FastAPI

MCP server

Streamlit UI

## SEVERITY LEGEND

Minor

Moderate

Major

Unknown

## ENGINEERING RIGOR



Combination-drug products are deliberately excluded, avoiding misattribution of interaction warnings to the wrong single ingredient.



Data normalization prevents duplicate drug-pair entries in the structured store.



Sentence-aware chunking preserves meaning across section boundaries in FDA label text.



Every API key is environment-managed — never hardcoded in source.



Ragas-based evaluation planned to measure faithfulness, precision, and recall of generated answers.

Python

PostgreSQL

SQLite

Qdrant

FastAPI

MCP

Streamlit

Claude API

openFDA

Evidence-grounded · Never a bare guess